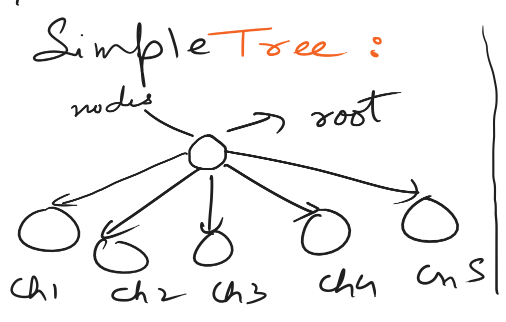
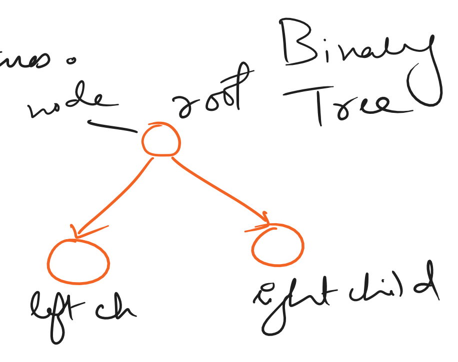
Binary Trees :→

Non-linear data structures.

Simple Tree :

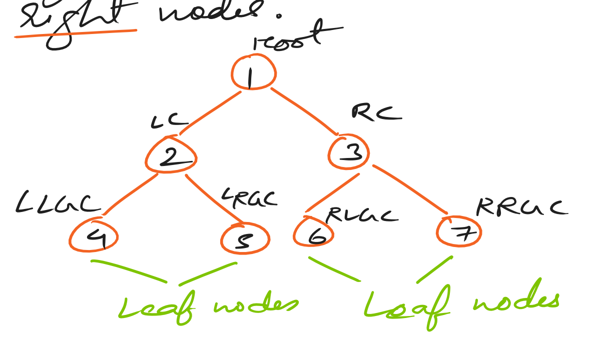


Binary Tree



A binary tree is a non-linear ds where each element is stored inside a node. Each node can have maximum 2 children or minimum 0 children.

The starting node of any tree is called the root node.
If any node doesn't have children, it is called a leaf node.
The two nodes of every node are called left & right nodes.



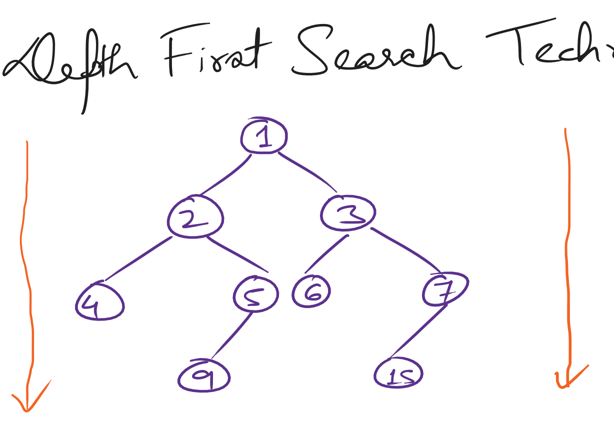
Trees

1. Trees
2. Binary Trees
3. Binary Search Trees
4. AVL Trees — Balanced BST
5. Heaps → Complete Binary Tree
6. Graphs

Traversal Techniques :

1. Depth first Search (DFS)
2. Breadth first Search (BFS)

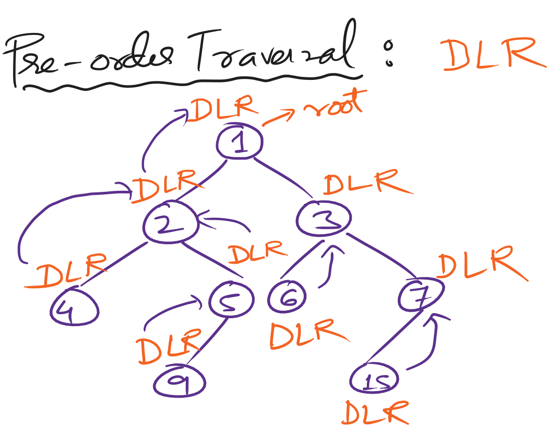
I. Depth First Search Technique



1. Pre Order Traversal DLR
2. In Order " LDR
3. Post Order " LRD

D → Data L → Left R → Right

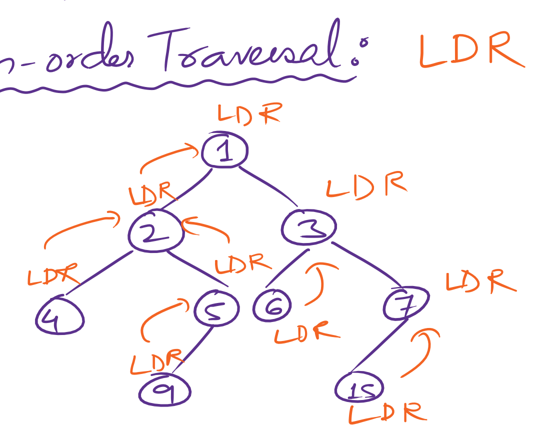
Pre-order Traversal : DLR



Output :→ 1, 2, 4, 5, 9, 3, 6, 7, 15

The root node data is printed at first.

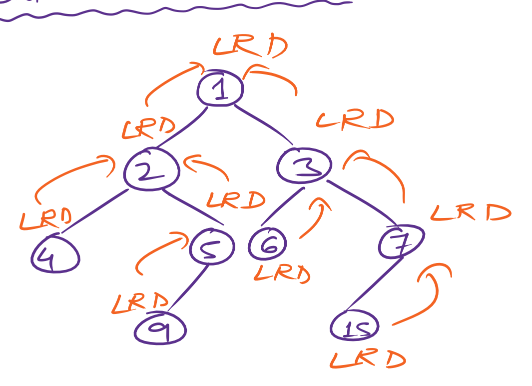
In-order Traversal : LDR



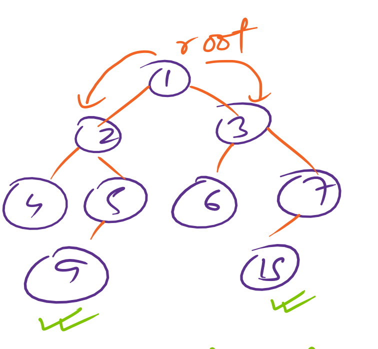
Output :→ 4, 2, 9, 5, 1, 6, 3, 15, 7

Post Order Traversal : LRD

Top to Bottom
Left to Right

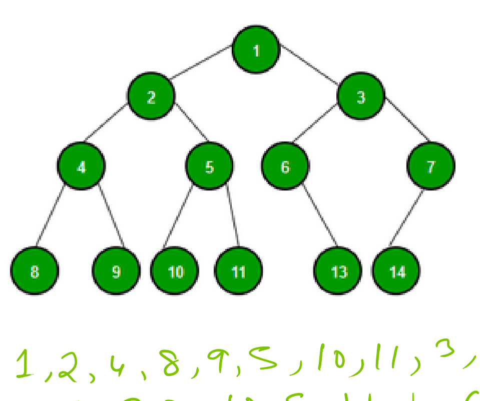


Output :→ 4, 9, 5, 2, 6, 15, 7, 3, 1



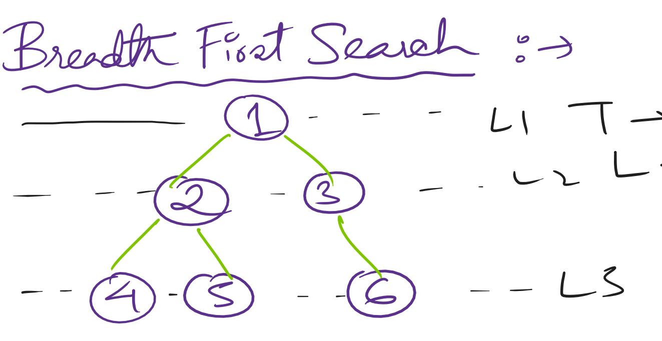
root. left. right. left,

Home work



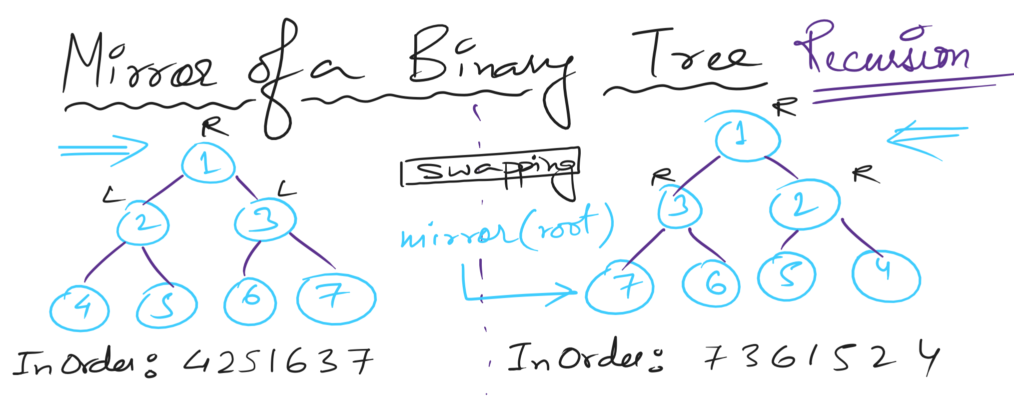
Pre → 1, 2, 4, 8, 9, 5, 10, 11, 3, 6, 13, 7, 14
In → 8, 4, 9, 2, 10, 5, 11, 1, 6, 13, 3, 14, 7
Post → 8, 9, 4, 10, 11, 5, 2, 13, 6, 14, 7, 3, 1

Breadth First Search :→



Level Order Traversal : 1, 2, 3, 4, 5, 6

Mirror of a Binary Tree Recursion



In Order: 4251637 In Order: 7361524

Identical Binary Trees